Constancy RoPE: An Empirical Study of Spectral Attention Gains at 4K

Working research report

October 17, 2026

Abstract

We test whether spectral feedback on rotary-position attention gains improves training-free 4K generation with FLUX.1-Krea-dev. Constancy RoPE compares the predicted clean latent spectrum with a native-resolution reference and applies a bounded, noise-gated correction to spatial attention gains. Two completed campaigns fail to demonstrate an improvement over the released SEGA baseline. In the second campaign, only SEGA plus time-matched constancy at gain $\eta = 1$ passes the eight-prompt development gate. On 187 held-out prompt pairs, its ImageReward difference is $+0.00857$ with a 95% confidence interval of $[-0.00841, 0.02742]$, while full-resolution MUSIQ changes by -0.20860 $[-0.70887, 0.31512]$. PickScore and crop MUSIQ decrease. These results do not support an efficacy or state-of-the-art claim. This working report records the completed experiments; it is not a submission-ready methods paper.

1 Question and tested controllers

The project asks whether feedback toward a model's native latent spectrum can improve resolution extrapolation through per-band attention gains. The target image size is 4096×4096, compared with a 1024×1024 native calibration size. The baseline is the released SEGA [11] allocation and RoPE base, identified in the implementation as `sega_official`. The released allocation differs from the paper-equation implementation; the two are not interchangeable baselines.

The first campaign tested the original constancy controller, then a time-matched native spectral target. The latter was also tested on SEGA's RoPE base with its reference temperature. This did not establish an improvement. The second campaign tested two effective-pool temperature rules, `alias_temp` and `alias_temp_sref`, and a composition of released SEGA with time-matched constancy, `sega_ctm`, at $\eta = 0.5$ and 1.0.

For the composed controller, let m_d be SEGA's released query/key multiplier for rotary pair d. A correction in spectral band b is

$$e_b = \tfrac{1}{2}\big(\log P_{\mathrm{native},b}(\sigma) + a - \log \widehat{P}_b\big), \tag{1}$$

$$\delta_b = \eta w_b \ \mathrm{clip}(e_b, -1, 1), \tag{2}$$

$$g_d = m_d^2 \exp\big(\mathrm{interp}(\delta)_d\big). \tag{3}$$

Here a is the anchor offset, w_b is the existing visibility gate, and $\widehat{P}_b$ is measured from the preceding transformer call's clean estimate. The native target is interpolated at the current scheduler noise level. Query and key are each multiplied by $m_d \exp(\delta_d/2)$. The first step has no correction, the index axis retains unit gain, and no additional global temperature is applied. Spectral powers use orthonormal FFTs and means within bands. The vision analogy motivates this controller; it does not establish its efficacy or novelty.

2 What the gain algebra establishes

Pair scaling and its causal limit. For fixed content vectors, the rotary-position logit [12] can be written as a finite trigonometric sum

$$\ell(\Delta) = \frac{1}{\sqrt{d_k}} \sum_j \mathrm{Re}\big[c_j e^{i\theta_j \Delta}\big]. \tag{4}$$

Applying a shared multiplier m_j to the query and key coordinates of pair j multiplies that term by m_j^2. Its sign and phase still depend on c_j. Increasing a high-frequency coefficient therefore does not by itself establish more local attention, higher image-band power, or sharper decoded detail. The three existing Krea development probes have inconsistent response signs: one prompt has no positive decoded-width diagonal among six tested bands. Those probes jointly perturb both spatial axes on NTK2, so they also cannot identify a single-axis response under released SEGA.

Standardized profile shape. Let ℓ be a nonconstant vector of sampled log powers and let $z(\ell) = (\ell - \operatorname{mean} \ell)/\operatorname{std}(\ell)$, using one fixed standard-deviation convention. For $\beta > 0$,

$$z(a\mathbf{1} + \beta\ell) = z(\ell). \tag{5}$$

This elementary identity applies to an ideal standardized-profile allocation. It does not imply invariance of SEGA's radial spread or its complete gain. The paper-inspired reference implementation adds a standard-deviation epsilon, and the released implementation floors energy and standard deviation. Their behavior is therefore invariant only under the corresponding inactive-floor or negligible-stabilizer conditions, up to floating-point error. The statement is about a formed log-power profile; raising a two-dimensional spectrum to a power need not commute with averaging it into that profile.

Allocation and scalar magnitude are coupled. Before clipping, write a centered SEGA multiplier as $m_j = m_{\mathrm{ref}}(1 + a_j)$ with $\operatorname{mean} a_j = 0$. Then

$$\operatorname{mean}(m_j^2) = m_{\mathrm{ref}}^2\big[1 + \operatorname{mean}(a_j^2)\big]. \tag{6}$$

Thus centering query/key multipliers does not preserve their mean squared magnitude, which is the mean coefficient gain. Clipping changes these moments again. The new RMS controls isolate this effect per spatial axis. They do not match the actual attention-logit distribution, its variance, or its entropy.

Forward-noise expectation and the gate. For the linear data–noise interpolation used to motivate rectified flow [7], assume independent data x_0 and unit white noise ε, with $x_\sigma = (1 - \sigma)x_0 + \sigma\varepsilon$. With an orthonormal transform, channel summation over C channels, and mean-per-band density on non-DC bands,

$$\mathbb{E}_\varepsilon P_b(x_\sigma) = (1 - \sigma)^2 P_b(x_0) + \sigma^2 C. \tag{7}$$

The cross term vanishes in expectation. This is not an exact decomposition of one noisy sample or one generated ODE trajectory. The controller's statistic $\rho_b = \max(P_b(x_\sigma)/(\sigma^2 C) - 1, 0)$ and gate $w_b = \rho_b/(1 + \rho_b)$ are design choices inspired by this identity. They do not estimate the uncertainty of the delayed clean prediction. Likewise, zero correction at an anchored target spectrum is an algebraic property of the rule, not a stability theorem for the full sampling process.

3 Executed protocol

All reported comparisons use the FLUX.1-Krea-dev transformer with FLUX.1-dev text encoders and VAE, 4096^2 output, 28 sampling steps, distilled guidance 4.5, and scheduler shift $\mu = 1.15$. The selected candidate and baseline share the executed schedule, seed convention, and SEGA RoPE base. Earlier FLUX.1-dev outputs exhibited a lattice artifact and are excluded from these results.

The 195 Aesthetic-4K evaluation captions are split into development rows 0–7 and held-out rows 8–194. Native calibration uses 16 separate Pick-a-Pic prompts. Development eligibility requires positive mean differences in both ImageReward and full-resolution MUSIQ, with CLIP difference at least -0.3. At most two variants may advance; no settings change after selection.

The second campaign reports exactly 187 matched seed-0 prompt pairs. Generation produced 188 images because row 7 was included to preserve task chunking; that row is excluded from all held-out tables. Confidence intervals use 2,000 paired bootstrap draws with bootstrap seed 0 and the 2.5 and 97.5 percentiles. Intervals are nominal per metric, without multiple-comparison adjustment. An interval crossing zero does not establish equivalence.

4 Results

4.1 Development selection

Table 1 records selection only. Both alias variants lose ImageReward. The $\eta = 0.5$ composition misses the MUSIQ gate by 0.007850. Only $\eta = 1$ advances; development scores are not generalization evidence.

Table 1: Development means on eight prompts, seed 0.

Variant	ImageReward	MUSIQ	CLIP	Advance
Released SEGA	1.423018	42.488060	30.320004	Reference
Alias temperature	1.281098	35.807356	29.366986	No
Alias, reference gain	1.277426	44.972845	30.238460	No
SEGA + constancy, $\eta = 0.5$	1.441334	42.480210	30.200250	No
SEGA + constancy, $\eta = 1$	1.448742	42.505928	30.342451	Yes

4.2 Held-out comparison

Table 2: Held-out seed-0 means and paired differences, candidate minus released SEGA, over 187 prompts. Higher is better except the last three rows.

Metric	SEGA	SEGA + constancy	Difference [95% CI]
ImageReward	1.18994	1.19852	$+0.00857$ $[-0.00841, +0.02742]$
PickScore	23.05255	22.97729	-0.07526 $[-0.10234, -0.04914]$
CLIP	29.13552	29.04160	-0.09392 $[-0.19651, -0.00368]$
MUSIQ, full image	45.22784	45.01924	-0.20860 $[-0.70887, +0.31512]$
MUSIQ, 1024 crops	58.32778	57.19612	-1.13166 $[-1.38754, -0.90219]$
CLIP-IQA, full image	0.60299	0.60593	$+0.00294$ $[-0.00018, +0.00609]$
CLIP-IQA, 1024 crops	0.58597	0.58846	$+0.00250$ $[-0.00070, +0.00563]$
Spectrum log RMSE	1.06838	1.06596	-0.00242 $[-0.00905, +0.00433]$
Spectrum slope error	0.19995	0.19741	-0.00254 $[-0.00678, +0.00192]$
Tiling at 1024	0.06836	0.06844	$+0.00008$ $[-0.00072, +0.00081]$

The selected composition has no demonstrated gain on ImageReward or full-resolution MUSIQ. PickScore, crop MUSIQ, and CLIP have negative nominal confidence intervals. The remaining intervals cross zero. This second campaign therefore does not repair the original controller's failure.

The earlier time-matched controller on SEGA's base also regresses on the same 187 seed-0 prompts. Full-resolution MUSIQ changes by -3.03602, with interval $[-4.28169, -1.81488]$. Crop MUSIQ changes by -2.19266, with interval $[-2.81241, -1.56666]$. Its ImageReward difference is unresolved at $+0.01782$ $[-0.02271, 0.05913]$. A temperature-only control on SEGA's base reduces full-resolution MUSIQ by 2.08550 points, with a paired interval $[-3.24526, -0.96407]$. These comparisons constrain the tested designs; they do not show that all spectral controllers must fail.

5 Limits and continuation

Novelty and current comparison scope. A successful continuation requires a distinct mechanism with competitive quality and cost against recent resolution-extrapolation methods. Adaptive spectral gains are already introduced by SEGA; temporal positional schedules by DyPE [4]; entropy-based concentration by UltraImage [16]; and text anchoring with dynamic temperature by TIDE [8]. HRDiT [15] combines spatial position alignment and head-adaptive pruning. More recently, AReA [10] changes attention from coarse cross-scale to progressively local receptive fields, while Spectral Correction Guidance [5] corrects sampling toward a reference spectrum. CSFlow [1] connects contrast sensitivity to flow-time weighting. In language models, DoPE [14] uses rotary-band Gram spectra and truncated

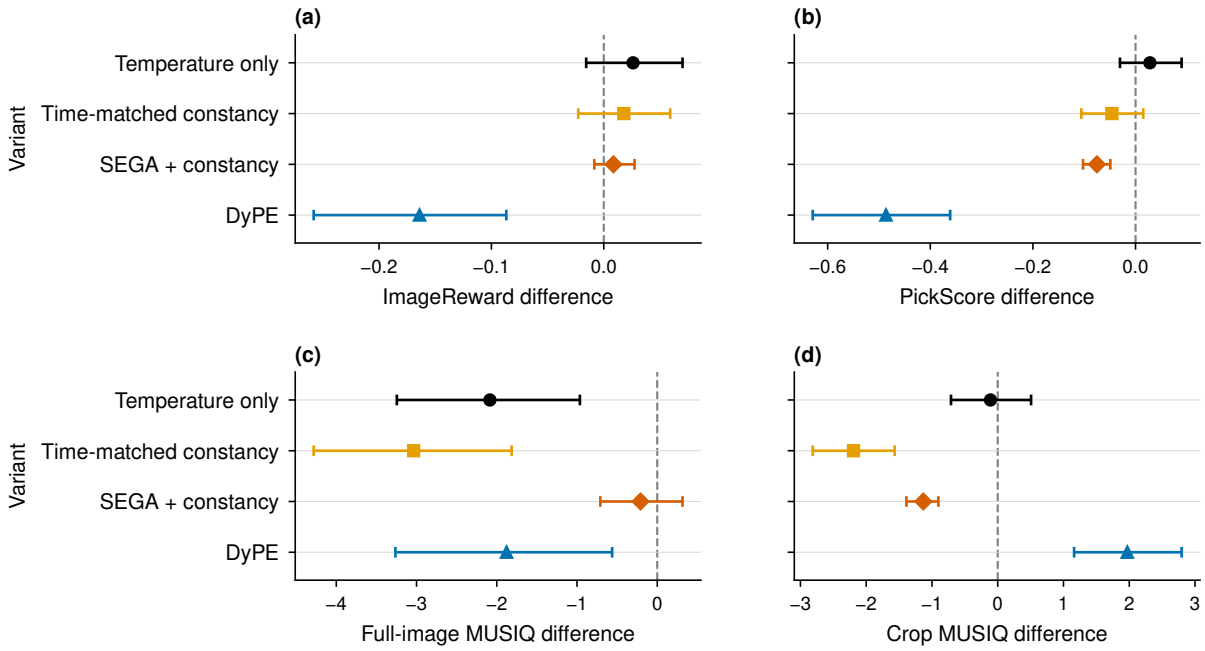


Figure 1: Neither constancy variant demonstrates a joint preference and quality gain over released SEGA. Points show paired differences; whiskers are nominal 95% prompt-bootstrap intervals from 2,000 draws over 187 prompts, single seed. Zero is the released SEGA baseline; higher is better in all panels. DyPE's crop quality gain accompanies a preference loss. The temperature-only and time-matched constancy variants use SEGA's RoPE base and reference temperature.

matrix entropy to select positional interventions. This is a prior-art boundary for covariance-based band suppression, not evidence of its efficacy in DiTs. Thus spectral feedback, perceptual motivation, normalization, or a timing change alone cannot establish the proposed contribution. The current RMS and timing controls are diagnostics. The completed SEGA comparisons do not establish competitiveness against this expanded set, and a development pass alone does not justify promoting a control to a new method. No near-SOTA or SOTA claim follows from the present evidence.

Local response diagnostic. The earlier octave interventions compare complete sampling trajectories, so they do not identify a local map from attention gain to spectral power. A new diagnostic replays a selected transformer state with identical latent, conditioning, and noise level. It perturbs height and width gains separately while preserving the affected axis's gain-vector RMS before attention-dtype quantization, alongside a spatial scalar control. This does not preserve query/key activation energy. Symmetric finite differences estimate a directional response of the clean-estimate spectrum along this normalized path, including BF16 quantization; smaller perturbations check prediction accuracy. Only the baseline output advances the sampler. CPU tests, including a small randomly initialized FLUX transformer, check exact trajectory and state restoration. Both pretrained 2048-resolution smoke tests reproduced the baseline latent endpoint bitwise and passed the same-state no-op replay, but failed the required exact gain-history comparison. The 4K stage therefore did not start, and no response result is accepted under the frozen protocol. A separate four-run repeatability diagnostic reproduced history drift between ordinary baselines. Across both prompts, BF16 gains, recorded intermediate states and velocities, and endpoints matched; the largest observed FP32 gain discrepancy was 1.19×10^{-7}. Repeating allocation on identical projected features also drifted under default arithmetic, while eight repeats with deterministic allocation matched for each prompt. This localizes a numerical repeatability issue to feature-to-gain arithmetic without identifying a unique kernel. A separately frozen diagnostic enables deterministic arithmetic only during allocation and records this numerical-policy change. Both 2048-resolution smoke tests and both 4K runs now pass the original exact replay, gain-history, and endpoint checks. These two smoke endpoints also match those from the original numerical policy. This is a bounded numerical validation, not evidence of general quality equivalence with the original baseline.

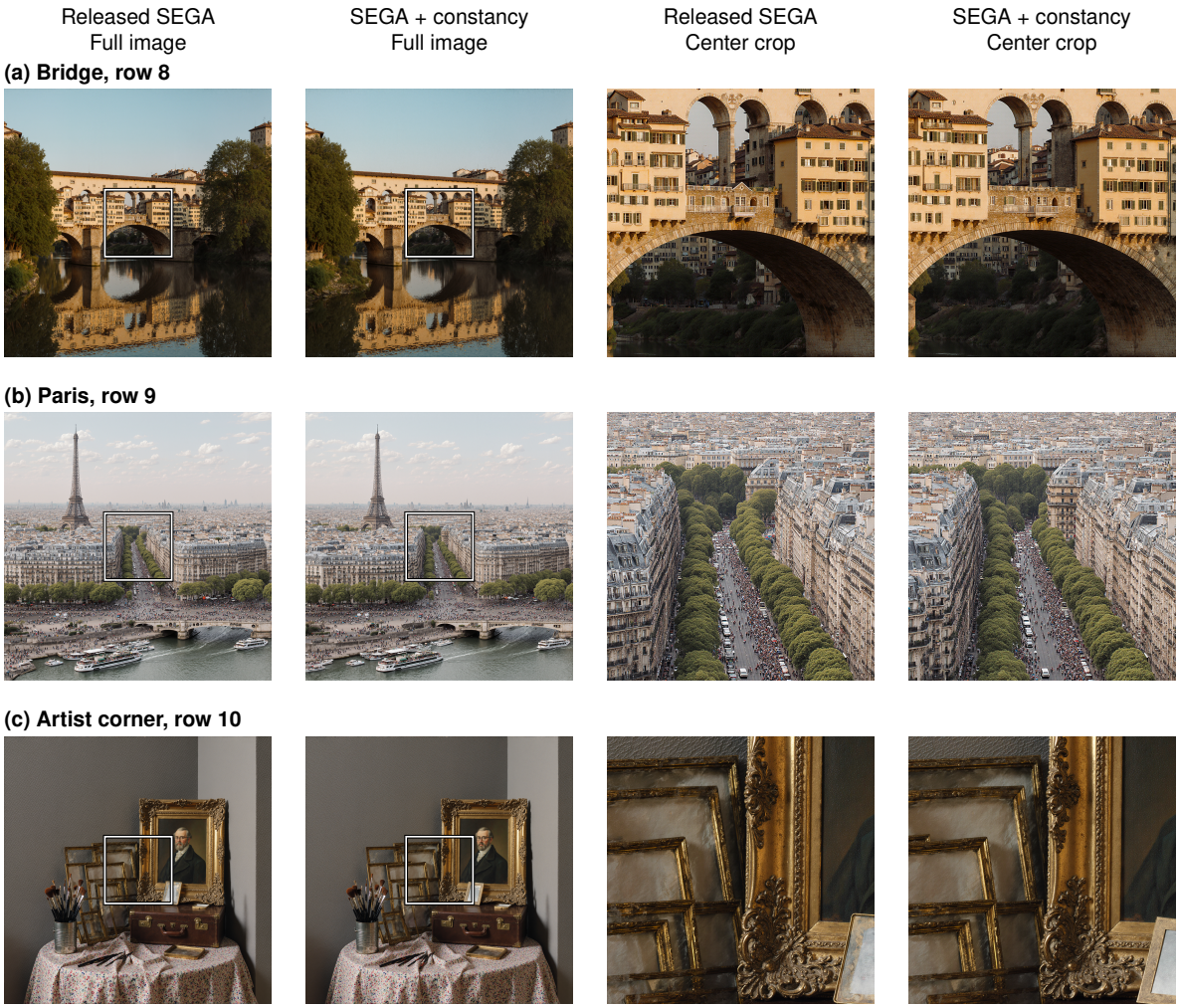


Figure 2: Fixed-order examples from the completed second campaign: the first three held-out captions, rows 8–10, seed 0. Columns compare released SEGA with SEGA plus constancy at $\eta = 1$, followed by their native 1024^2 center crops. Boxes show the crop coordinates in the 4096^2 full outputs. The same pixel coordinates are used in each image; generated content need not align exactly. Full images are reduced for display, and crops receive no enhancement. Rows were fixed before viewing these images and were not selected by scores. Three examples cannot establish preference or quality differences; the paired 187-prompt measurements supply the quantitative evidence.

On two development prompts, we probe steps 6, 14, and 22 of 28 with full perturbation magnitude 0.10 and unfit half magnitude 0.05. For each spatial axis, the selected rotary octave is -11; it lies below the minimum nonzero spatial DFT frequency of the token grid. These measurements therefore do not characterize all rotary bands. Prediction error is the norm of the combined positive and negative half-step residuals divided by the corresponding observed response norm; a value of one matches a zero-response predictor. Across the 12 axis-octave cases, the median error is 0.937 (range 0.709–1.354), with nine cases below one. Across six spatial-scalar cases, the median is 0.426 (range 0.050–3.039). The late scalar errors are 0.104 and 0.050, but the early failure on one prompt prevents a uniform predictability claim. Raw response magnitudes are not directly comparable because the actuator norms differ. These results do not support inverting the current octave response map. An unprespecified secondary analysis uses the actual BF16 log-gain changes as control coordinates. It fits only full-magnitude observations and predicts the untouched half-magnitude responses. Axial errors remain 0.688–1.339 and scalar errors 0.029–3.004. No intervention vanishes entirely through quantization. However, 31–49% of each axial half-step gain-change norm lies outside the full-step secant direction (11–16% for scalar controls). A one-dimensional coordinate correction therefore does not separate quantization from nonlinear response; it does not change the frozen diagnostic's acceptance rules. The diagnostic adds 39 transformer calls per 4K trajectory; recorded baseline and probed sampling times are 201/485 and 185/441 seconds for the two prompts. This is measurement overhead, not an efficient generation method.

This is an instrument for testing the controller hypothesis, not a new validated generation method. SCG acts directly on latent Fourier magnitudes, whereas this diagnostic measures the response to internal attention gains [5, 11]; this difference alone does not establish novelty or improved quality. Jacobian-based analysis and correction are established: Graikos et al. study denoiser-input Jacobian spectra and train with response regularization, while JAPS uses denoiser Jacobian information for inverse-problem guidance [2, 3]. Our measured object is instead the response of Fourier-power statistics to attention gains at a fixed state. Attention Frequency Modulation already edits spatial frequency bands of cross-attention logits and uses entropy-dependent gating [9]. Thus spectral attention control itself is also established. The proposed response map remains a hypothesis to test, not sufficient evidence of a novel controller. Local prediction must also transfer to the final image at acceptable sampling cost. The failed two-prompt smoke test is retained as an implementation finding, not a result about generation quality or the proposed response model.

Observer invariance diagnostic. A separate source audit identifies a limitation of the released SEGA observer. For the standard text-conditioned FLUX path, with fixed attention gains and no optional adapters, each residual branch reads a LayerNorm-normalized state and the output projection follows a final LayerNorm. In real arithmetic, $\mathrm{LN}(h_i + a_i\mathbf{1}) = \mathrm{LN}(h_i)$ for a scalar a_i at each token. Residual additions preserve this common-channel shift until the final normalization removes it. Such residual-stream centering is established in transformer analysis [13]. However, SEGA takes the spatial spectrum of the channel mean of the initial projected features. That mean changes by a_i, so the observer can change even when the fixed-gain denoiser function does not. This argument concerns a representation dependence, not proof of poor quality on the released weights. Finite-precision reparameterizations require separate checks. On a randomly initialized miniature FLUX with one joint and one single block, three common-channel reparameterizations preserve the output within 1.23×10^{-15} maximum absolute error while changing or zeroing the observer's spatial energy. A channel-dependent negative control preserves the channel mean but changes the output by 2.025 maximum absolute error. Parameters and outputs are FP64; the installed rotary implementation includes FP32 casts. Seven CPU checks pass. This is an architecture-level numerical counterexample, with no pretrained, BF16, trajectory, or image-quality result. An independent miniature test uses the released SEGA observer and the installed gain processor on a 4×128 token grid. A spatially varying common-channel intervention changes gains by 0.1175 and the adaptive output by 0.00717 maximum absolute difference. Keeping original gains reduces the output difference to 1.34×10^{-15}; applying the changed gains to original features reproduces the shifted adaptive output to the same tolerance. Seven additional checks pass. This isolates gain mediation in the miniature; the intervention is on activations at one input, not a universal parameter reparameterization.

A separately frozen pretrained BF16 audit evaluates two existing development prompts at 4096^2. Each input is an artificial mixture of a validated sampling endpoint and fixed CPU-generated Gaussian noise at $\sigma = 0.5$, rather than a recorded reverse-trajectory state. The intervention replaces the projected channel-mean map with a fixed spatial harmonic while matching its original mean and population standard deviation before BF16 rounding. The original and shifted features are crossed with their respective SEGA gains; one additional original-state repeat gives five forwards per prompt, ten in total. Saved original/repeat outputs match byte for byte, and provenance, gain crossings, saved tensors, and output contrasts pass independent CPU validation.

Let F_{ab} denote the velocity with original ($a = 0$) or shifted ($a = 1$) features and original ($b = 0$) or shifted-observer ($b = 1$) gains. The gain-only effect $\|F_{01} - F_{00}\|_2$ is 109.986 and 66.181 for the two prompts, compared with fixed-gain residuals $\|F_{10} - F_{00}\|_2$ of 16.505 and 16.448. Their ratios are 6.664 and 4.024. The adaptive changes $F_{11} - F_{00}$ align with the gain-only changes with cosines 0.988 and 0.963. However, $\|F_{11} - F_{01}\|_2/\|F_{01} - F_{00}\|_2$ remains 0.156 and 0.271, and the interaction residual has relative magnitude 0.215 and 0.366 against the gain-only effect. Thus the pretrained intervention has a substantial gain-mediated component, with non-negligible finite-precision or architectural residuals; it does not reproduce exact mathematical invariance. Full projected features and conditioning were not saved, so their checks remain runtime receipts. These two synthetic-state measurements do not establish a defect on natural trajectories, an image-quality effect, or a new method.

Changing the observer to channel-centered spectral energy would remove this particular dependence, but would not itself establish novelty or quality. AReA already rejects channel means because opposite-sign feature changes can cancel, and uses channel standard deviation before its wavelet detail measurement [10]. Its statistic is also invariant to a common-channel shift in real arithmetic. The SEGA-specific audit establishes sensitivity under the declared intervention, not a new-method claim.

Spatial-complement attention diagnostic. A separate passive capture tests whether spatial complement statistics justify further method development. PISA already formulates a first-order correction using the mean within-block value–key cross-moment over unselected blocks, then replaces it with an all-block mean for efficiency [6]. Computing the conditional mean is therefore a refinement of a known expression, not a new attention approximation.

We reuse the two artificial inputs above with their original gains. One unmodified forward and one capture forward per prompt give four forwards in total; their final outputs match byte for byte. We retain the actual BF16 post-gain queries, keys, values and attention outputs at calls 0, 18 and 56, heads 0 and 12, and 16 fixed image queries. Independent FP64 evaluation uses these saved tensors and the same dense softmax target. Image tokens form 16×16 cells; the exact set contains cells within Chebyshev distance two of the query cell and all text tokens. All remaining cells share a centroid mass approximation. We compare zeroth-order values, global or complementary mean cross-moments, and individually weighted cell cross-moments.

Across all 192 fixed cases, median relative output errors are 0.3133 for exact sparse attention, 0.0400 for zeroth-order completion, 0.0203 for the global first-order mean, 0.0192 for the complementary mean, and 0.0116 for the weighted cell correction. The corresponding maxima are 0.9297, 0.6063, 0.5930, 0.5940, and 0.5349. These are descriptive summaries of this stencil, not a representative benchmark or a reproduction of PISA's routing and fused kernels. The complementary mean improves 138 cases and worsens 54. Its absolute error change exceeds the separately measured BF16-versus-FP64 attention gap in only 33 cases: 26 improvements and seven regressions. Errors remain large in the sampled final joint block, including with individually weighted moments. Independent blockwise calculations agree with all reported outputs within 3.91×10^{-14}. The small, mixed conditional-mean benefit does not justify a GPU implementation or a generation campaign for this branch. No image-quality, resolution-extrapolation, speed, or novelty claim follows.

The second-campaign candidate has one held-out seed and one evaluated backbone at one resolution. No second-seed replication of this candidate, new backbone, human preference study, or broader resolution claim is supported. Aggregate FID/KID scores from the raw 188-image evaluation include development row 7 and are excluded from the held-out conclusion. No new throughput comparison is claimed in this report.

The earlier extended theory and literature sections remain in the source tree but are excluded because they describe superseded settings and unresolved claims. Section 2 retains only qualified algebraic observations. The third diagnostic below tests gain magnitude, frequency allocation, and controller timing under a frozen development-only protocol. The already inspected held-out set must not become a tuning set presented as fresh evaluation.

Third development diagnostic. The three prespecified controls and a fresh released-SEGA baseline completed on eight development prompts, seed 0. These are exploratory selection scores, not held-out efficacy evidence.

PickScore differences against the same baseline are RMS-matched uniform: -0.0832; RMS-normalized allocation: -0.0666; Lag-aligned constancy: -0.0546. RMS-normalized allocation passed the development gate. These controls remain diagnostics, not an established novel method. A separate set of 64 confirmation captions was frozen before these outcomes; its use requires a distinct method and a recent-baseline protocol. No confirmation result or efficacy claim follows from this table.

Table 3: Third development diagnostic. Differences are against the fresh same-hardware SEGA baseline.

Variant	ΔIR	ΔMUSIQ	ΔCLIP	Δcrop MUSIQ	Gate
RMS-matched uniform	-0.0174	-1.5667	+0.5868	-0.5177	Fail
RMS-normalized allocation	+0.0363	+0.3889	-0.0655	-0.2150	Pass
Lag-aligned constancy	+0.0193	-0.2233	+0.0092	-0.1530	Fail

Conditional confirmation protocol. Before the third development outcome, a separate set of 64 captions was frozen from Aesthetic-4K's 2048-source split after excluding normalized matches to all prior evaluation and calibration captions. Fixed hash ordering selected the captions without inspecting generation or scores. A candidate that satisfies both the method-novelty review and a separately frozen development protocol would use its unchanged settings at 4096^2 with a fresh SEGA baseline. The captions are fresh to this project's evaluation, with no model-training disjointness or semantic-near-duplicate guarantee. They do not supply matched 4K reference images. The prespecified analysis uses 10,000 paired prompt-bootstrap draws, seed 0, and common resampling indices across metrics and candidates. Nominal 95% intervals accompany all metrics. For the two primary metrics, ImageReward and full-image MUSIQ, Bonferroni-adjusted percentile intervals have coverage $1-0.05/(2K)$ for K selected candidates. Joint primary support requires both adjusted lower bounds to exceed zero. These are approximate bootstrap intervals, not exact finite-sample guarantees. Secondary losses remain part of the report; confirmation cannot select further controller settings. No confirmation result is available in this snapshot.

Evidence provenance. The completed campaign is recorded at implementation commit `a1ede1a` on October 16, 2026. Its archived paired table, per-prompt scores, and analysis protocol are in `.agent/notes/main2/`. The current held-out table is generated from those CSVs after checking its means and paired intervals against the per-prompt records. This check reuses saved scorer outputs; it is not a fresh image-scoring reproduction. The previous session records 194 passing CPU tests. The third diagnostic adds new inference and evaluation; its development-only outcome is reported separately above.

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